Resource cost management

Project Name: Smart parking platform

Course: CIS 4374 – Software project management

Date : 10/15/26

Part 1: Roles, Resources & Allocations

Role	Resource Name	Skill Level Assumption	% Allocation (FTE)	Primary Responsibilities
Project Manager / Scrum Master	Alex Morgan	Senior (PMP / Agile Certified)	50% (0.50 FTE)	Sprint facilitation, backlog health, risk tracking, stakeholder reporting[cite: 14, 16]
Backend & Cloud Engineer	David Chen	Mid-Level (Node.js, Express, AWS EC2/RDS)	50% (0.50 FTE)	REST APIs, JWT auth, database schemas, payment tokenization[cite: 15]
Frontend / Mobile Engineer	Sarah Jenkins	Mid-Level (Vue.js, Mobile UI/UX, Map APIs)	50% (0.50 FTE)	Driver reservation interfaces, GPS map integration, operator portal[cite: 14, 15]
QA & Security Specialist	Marcus Vance	Junior / Mid-Level (Jest, Postman, AppSec)	50% (0.50 FTE)	API test suites, automated regression, vulnerability scans, access control[cite: 15]
Database & DevOps Architect	Elena Rostova	Senior (IaC, PostgreSQL, Docker, Redis)	25% (0.25 FTE)	CI/CD pipelines, container orchestration, caching layers, high-availability[cite: 15]

Part 2: Resource Plan, Cost Plan & Budgeted Monthly Cashflow

2A. Bottom-Up Labor Resource Plan

Role	Name	FTE	Estimated Hours	Base Hourly Rate	Total Cost
Project Manager	Alex Morgan	$1	$240	$55	$13,200
Backend Engineer	David Chen	$1	$240	$50	$12,000
Frontend Engineer	Sarah Jenkins	$1	$240	$50	$12,000
QA & Security Specialist	Marcus Vance	$1	$240	$40	$9,600
Database / DevOps Architect	Elena Rostova	$0	$120	$65	$7,800
TOTAL LABOR			1,080		$54,600

2B. Bottom-Up Project Cost Plan / Non-Labor Budget

Cost Category	Item Description	Unit Cost	Quantity / Duration	Total Cost	Cost Basis / Assumption
Human Resources	Project Engineering & PM Labor	$54,600	16-Week Lifecycle	$54,600	Bottom-up estimation from Resource Plan (1,080 hrs)
Cloud Hosting	AWS Infrastructure (EC2, RDS, S3, VPC)	$150 / mo	4 Months	$600	Estimated staging/production hosting footprint
SaaS & Dev Tools	GitHub Team, Jira/Trello, Postman	$40 / mo	4 Months	$160	Standard developer collaboration software licenses[cite: 14, 15]
Third-Party APIs	Google Maps Platform, Stripe Sandbox, Twilio	$80 / mo	4 Months	$320	Geocoding, SMS OTP, and transaction gateway test charges[cite: 15]
Hardware Prototyping	IoT Gate Controller Kit (Raspberry Pi & RFID)	$180 lump	1 Kit	$180	Off-the-shelf barrier simulator equipment
Direct Cost Subtotal				$56,060	Total direct planned expenditures

Contingency Reserve	Technical & schedule risk buffer (10%)	10% of Direct	Lump Sum	$5,606	Mitigates race condition & API integration risks[cite: 15]
Management Reserve	Unplanned scope & compliance changes (5%)	5% of Direct	Lump Sum	$2,803	Standard project management governance buffer
TOTAL PROJECT BUDGET				$64,469	Total baseline budget

2C. Budgeted Monthly Cashflow (4-Month Project Lifecycle)

Expense Category	Month 1 (Initiation & Spikes)	Month 2 (Core Sprint Dev)	Month 3 (Integration & QA)	Month 4 (Launch & Final Review)	Total ($)
Labor Expenses	$13,650	$13,650	$13,650	$13,650	$54,600
Cloud & Infrastructure	$150	$150	$150	$150	$600
SaaS & Developer Tools	$40	$40	$40	$40	$160
Third-Party APIs	$40	$80	$120	$80	$320
Hardware Simulation Kits	$180	$0	$0	$0	$180
Direct Monthly Outflow	$14,060	$13,920	13,960	$13,920	$56,060
Cumulative Cashflow	$14,060	$27,980	$41,940	$56,060	$56,060

Part 3: Project RACI Matrix

- **R** = Responsible (The person who completes the work)
- **A** = Accountable (The single person who approves and owns the outcome)
- **C** = Consulted (Provides two-way input and subject-matter expertise)
- **I** = Informed (Kept updated on progress and results)

Deliverable / Activity	Project Manager	Backend Dev	Frontend Dev	QA / Security	DevOps Architect
Sprint Backlog & User Story Prioritization	A	C	C	I	I
System Architecture & DB Schema Design	I	R	C	C	A
JWT Authentication & Role-Based Access (RBAC)	I	R	C	A	C
Driver GPS Map & Live Availability UI	I	C	R	I	I
Gate Barrier IoT & Telemetry API Integration	I	R	I	C	A
Payment Gateway & Reservation Locking Engine	I	R	C	C	I
Automated CI/CD Pipeline & AWS Deployment	I	C	I	I	R
Security Hardening & Penetration Testing	I	C	C	R	A
Sprint Review, Demo & Stakeholder Sign-Off	A	C	C	C	I